VOLLEY | Final-review evidence summary

23 October 2026

Adityavardhan Mishra · Pratham Chawla | Symbiosis Institute of Technology, Pune

Research question. Can a shared, commandable CubeSat deployer provide useful release states while meeting the installed mass and integration burden of a 12 × 3U reference case?

Decision. The evaluated Gen5 configuration is **not selected** for that case. The computational study records both conditional orbital benefit and adverse system findings. No VOLLEY hardware was built or tested.

Finding	Result and evidence boundary
Release speed	16.029 m/s belongs to a historical periodic-force model. P118's finite 3-D **analytic** force screen gives 12.448 m/s under ideal phase and omitted circuit losses. P121 independently reproduces 1.042 kJ ideal full-depth work under shared inputs; a P122 illustrative 14 mm gap gives 0.907 kJ. No selected rating exists.
Electrical chain	Historical gross draw is 2,782 J; its 124.488 J compact-output remainder resolves to assumed converter, auxiliary and numerical terms. A conditional finite-force full-winding bank rerun gives 2,099 J gross at an unselected 12 mΩ source. P125 finds that an older 116–185 mΩ single-string ESR band fails source power before release; ideal parallel strings add cells and mass. No selected drive rating exists.
Installed mass	Modeled dry mass is 126.6 kg, or 10.55 kg per 3U customer, against an approximate 6 kg spring-canister comparator. The predeclared 15% parity band fails.
CAD fit	The reference side-fed layout needs 537 mm within 526 mm clear width and overlaps the track by 32,915 mm³ per cassette. The separate widened R1 geometry clears twelve scripted routes; it lacks a selected feeder, retention and revised mass.
Orbital value	+28.8008 km immediate two-body axis rise is checked **for an assumed historical 16.029 m/s release**, not as achieved Gen5 behavior. The 1.60× atmosphere-dependent lifetime figure remains open.
Matched mission	A single assumed target yields ideal host propellant of 2.693 kg spring versus 0.827 kg at the finite Gen5 *ideal upper* speed. No bounded twelve-shot campaign closes; this is not a proven service advantage.

Selected feeder. FM08-R3 adopts independent guided lead-screw cassettes, fixed 1U spacers, capped-channel transfer and one indexed receiver. Its selected-design package provides the geometry, calculations and executable sequence checks. This is a computational academic-design selection, with hardware qualification reserved for later tests.

Evidence classes. The pack contains source scripts, captured run outputs, numerical cross-checks, FreeCAD native review assemblies, STEP exports, figures, reports and an explicit claim-to-source map. These establish only their stated models, geometry and boundary conditions. They do not establish payload qualification, supplier component ratings, host/provider approval, customer demand or flight readiness.

Next gate. Independently check finite integrated thrust; rerun the coupled circuit and trajectory; use the selected FM08-R3 feeder design as the basis for instrumented physical development and reconcile its full launcher integration. The later Gen6 P130 surrogate lets both commanded release and spring-plus-host meet four ideal targets, but assumed device-plus-fuel mass favors springs by 10.7177 kg. P131–P132 provide two partial FreeCAD/STEP first-article envelopes and necessary energy bounds;

neither candidate is build ready. An instrumented article follows only after G0/G1 interface, source, contact, brake and installed-system review. The 1 km/s line is long-range research, not a result.

Source trail. `P118_gen5_finite_force_map.md`; `P119_gen5_finite_force_fem2d.md`; `P120_gen5_finite_coupled_shot.md`; `P121_gen5_finite_force_surface3d.md`; `P122_gen5_finite_force_sensitivity.md`; `P117_rated_energy_mass_audit.md`; `P116_gen5_assembly_packaging.md`; `P115_rated_orbit_cartesian.md`; `GEN5_COMPUTATIONAL_REVIEW.pdf`; `GEN5_CAD_REVIEW.pdf`; `CLAIM_EVIDENCE_MAP.md`. The complete pack is in the VOLLEY-thesis repository's `university/final_revi` directory.